

Blackbodies

A beam of light falls on an object

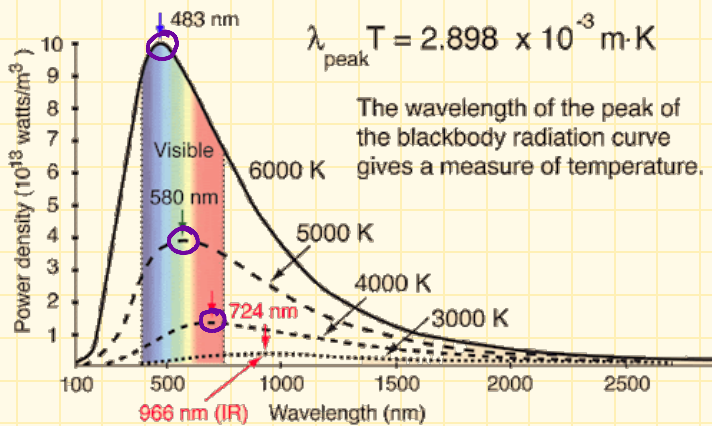
- reflected
- transmitted
- absorbed.

Blackbody → an object that absorbs all incoming radiation, at every wavelength.

If something is perfect at absorbing, it must also be perfect at emitting.
why? Because of thermal equilibrium.

If it absorbed energy but didn't emit it → heat up infinitely.

So a blackbody is the most efficient absorber and the most efficient emitter of radiation.



Hotter objects emit peak radiation at shorter wavelengths.

low temperature → peak infrared (long wavelength) → appear red / dull.

Medium " → " visible → yellow / white.

High " → " blue / UV (short λ)

Temperature is a measure of how much energy is stored in microscopic motion

so when temperature increases



Particles move more violently



Accelerations increase

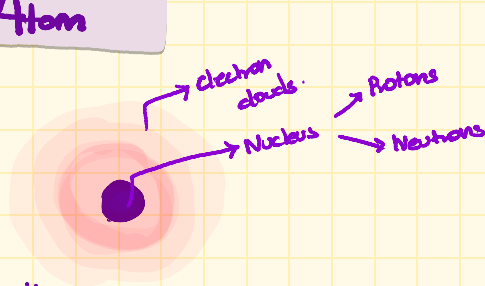


Radiation becomes stronger and shifts in character.

Blackbody would absorb all incoming radiation, it reaches perfect thermal equilibrium and it re-emits radiation purely based on its temp.

Structure of an Atom

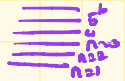
Electrons → energy states



The force holding it together:

Electromagnetic Attraction

- protons → positive
- electrons → negative.



Quantization

Electrons can only exist in specific energy levels.

- ground states ($n=1$)
- excited states ($n=2, 3, 4$)

There is no continuous energy - only discrete levels. (electron orbits)

Energy transitions

If an electron moves up → it absorbs energy
moves down → it releases energy.

And that released energy becomes $\rightarrow$ a photon (light)

When an electron drops between levels:

$$E = hf = \frac{hc}{\lambda}$$

The energy difference between levels determines the wavelength of light.

Blackbody radiation

- energy is continuous
- spectrum is smooth curve

Collective motion of many particles
 $\rightarrow$ atoms vibrating
 $\rightarrow$ charges acc.
 $\rightarrow$ energy spread continuously.

Atomic emission

- energy is discrete
- spectrum is sharp lines.

$\rightarrow$ Individual or transitions
 $\rightarrow$ specific jumps
 $\rightarrow$ fixed energy difference.

Conservation Of Energy

Energy cannot disappear or appear, it must transform exactly.

$$\Delta E = hf = \frac{hc}{\lambda}$$

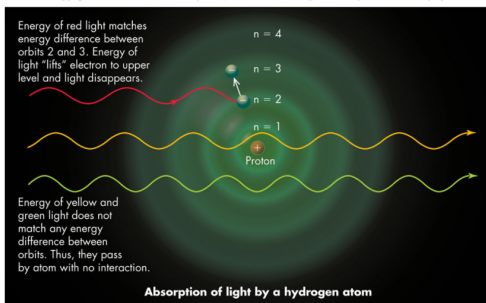
Only photons with the exact right energy can interact with an atom.

Red photon $\rightarrow$ matches energy gap $\rightarrow$ absorbed.

Yellow / green photons
 $\rightarrow$ don't match $\rightarrow$ pass through.

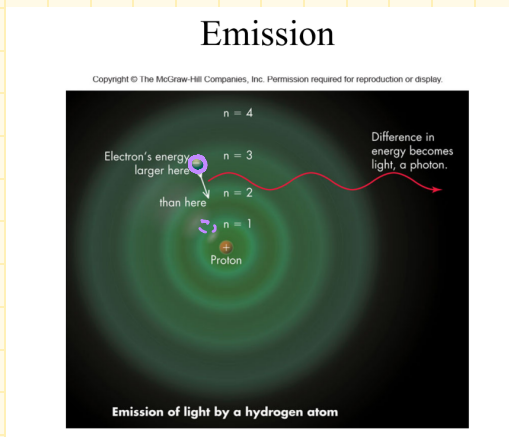
Absorption

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An e^- drops from a higher level to a lower one. It must release energy

The energy becomes a photon with a very specific λ .



What happens if too many atoms are being emitted?

- each transition produces a specific λ .
- you get a set of bright lines.

This is an emission spectrum.

Absorption Spectrum

If white light passes through a gas:

- certain wavelengths are absorbed
- those exact wavelength disappear.

You see a continuous spectrum with dark lines missing.

Spectra

Blackbody $\rightarrow$ continuous spectrum

Atoms $\rightarrow$ discrete lines.

Real system: (stars)

- The hot dense interior: blackbody radiation
- The cooler outer gas: absorb specific wavelengths.



A continuous spectrum with dark absorption lines.

Spectroscopy

You take light $\rightarrow$ spread it into $\lambda \rightarrow$ analyse it.

Why does it work?

Each atom has unique energy levels. So it absorbs/emits unique λ .

Types Of Spectrum

They arise from how matter interacts with light.

1. Continuous spectrum
2. Emission line spectrum
3. Absorption line spectrum.